

IAN HAMBLIN

GRAPHICS / RENDERING ENGINEER

(WEBGL + UNITY)

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const GRAPHICS & RENDERING = [
  { skill: 'Unity SRP (URP/HDRP)', years: 4 },
  { skill: 'ShaderLab / HLSL', years: 5 },
  { skill: 'Shader Graph', years: 2 },
  { skill: 'Three.js', years: 7 },
  { skill: 'WebGL', years: 7 },
  { skill: 'Shaders (GLSL)', years: 5 },
  { skill: 'React Three Fiber', years: 3 },
  { skill: 'GPU Instancing / Batching / LOD',
    years: 4 },
]

const LANGUAGES & FRAMEWORKS = [
  { skill: 'TypeScript', years: 9 },
  { skill: 'JavaScript', years: 20 },
  { skill: 'C# (Unity)', years: 6 },
  { skill: 'React', years: 5 },
  { skill: 'Node.js', years: 5 },
]

const 3D DCC & ENGINES = [
  { skill: 'Blender', years: 5 },
  { skill: '3ds Max', years: 15 },
  { skill: 'Unity', years: 6 },
  { skill: 'Unreal Engine', years: 4 },
  { skill: 'Substance Painter', years: 3 },
]

const TOOLS & DEVOPS = [
  { skill: 'Git', years: 10 },
  { skill: 'CI/CD', years: 4 },
  { skill: 'Docker', years: 3 },
  { skill: 'AWS', years: 5 },
]
```

EDUCATION

2004	BA Hons in Creative Visualisation, University of Teesside
1999	A Levels: Maths, Physics, Technology — St John Fisher Sixth Form

ABOUT ME

Graphics-focused engineer blending 3D art foundations with modern WebGL/Three.js and Unity rendering. Experienced shipping realtime visualisation tools and data-driven rendering for large geospatial and simulation datasets.

Comfortable across shader authoring (ShaderLab/HLSL, GLSL), Unity SRP (URP/HDRP), performance tuning, and building clean UI for technical users. Enjoys collaborating with artists and engineers to deliver visually impressive, performant, and maintainable experiences.

EMPLOYMENT HISTORY

3DEO – Lead Visual Engineer	2016 – 2024
- Built a browser-based realtime 3D terrain visualizer (TypeScript + Three.js + WebGL). - Designed colour-mapping and shader pipelines to analyse bathymetry datasets at scale. - Reduced volumetric calc times from ~30s to 0.1s with GPU accelerated techniques. - Implemented temporal data blending and elevation deltas using custom GLSL shaders. - Delivered a realtime 3D flythrough showcased at COP26; integrated video and interactions. - Worked across AWS (EC2, S3) and web stack to ship performant visual tools.	
Freelance – Graphics Engineer / 3D	2011 – 2016 & 2024 –
Generalist	Present

- Unity C# graphics: authored shaders (ShaderLab/HLSL), configured URP/HDRP, and tuned lighting/post-processing.
- Developed Three.js based editors and realtime viewers; implemented tiling and material workflows.
- Created Python + Blender add-ons for streamlined asset export/import pipelines.
- Built VR driving simulator and multiple AR experiences (Unity + Web); integrated live sensor/geo data.
- Implemented WebRTC pipeline from browser to Blender for high-quality realtime rendering.
- Produced marketing CG renders (e.g., M-Sport Raptor Dakar); modelled complex assets.

HIGHLIGHTS

- Unity graphics pipeline: URP/HDRP setup, lighting, post-processing, render queues.
- Shader authoring: ShaderLab/HLSL (Unity) and GLSL (WebGL); surface/fragment shaders.
- Performance: tuned large scenes with LOD, GPU instancing, batching, and frustum culling.
- UX: built technical analysis tools (cross-sections, delta views) for domain experts.
- Pipelines: Blender/3ds Max tools; automated ingest and material workflows.

Caspian Learning – Artist / Developer

2005 – 2015

- Shipped education/training titles; authored environments, characters, and animation.
- Collaborated on rigging and tooling to improve runtime performance and iteration speed.
- Introduced lightmapping workflows; built custom 3ds Max material editor for pipeline wins.